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Recognised Standard 08

Conduct of mine emergency exercises

Coal Mining Safety and Health Act 1999

Resources Safety & Health Queensland

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Recognised Standards may be updated from time to time. To ensure you have the latest version, check the website: https://www.business.qld.gov.au/industries/mining-energy-water/resources/safety-health/mining/legislation-standards or contact your local office.

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Recognised standards

This document is issued in accordance with PART 5—RECOGNISED STANDARDS and section 37(3) of the *Coal Mining Safety and Health Act 1999*.

PART 5 - RECOGNISED STANDARDS

71 Purpose of recognised standards

A standard may be made for safety and health (a "recognised standard") stating ways to achieve an acceptable level of risk to persons arising out of coal mining operations.

72 Recognised standards

(1) The Minister may make recognised standards.

(2) The Minister must notify the making of a recognised standard by gazette notice.

(3) The CEO must publish on a Queensland government website each recognised standard and any document applied, adopted or incorporated by the standard.

(4) In this section—

Queensland government website means a website with a URL that contains 'qld.gov.au', other than the website of a local government

73 Use of recognised standards in proceedings

A recognised standard is admissible in evidence in a proceeding if—

(a) the proceeding relates to a contravention of a safety and health obligation imposed on a person under part 3; and

(b) it is claimed that the person contravened the obligation by failing to achieve an acceptable level of risk; and

(c) the recognised standard is about achieving an acceptable level of risk.

PART 3 - SAFETY AND HEALTH OBLIGATIONS

37 How obligation can be discharged if regulation or recognised standard made

(3) …. if a recognised standard states a way or ways of achieving an acceptable level of risk, a person discharges the person's safety and health obligation in relation to the risk only by—

(a) adopting and following a stated way; or

(b) adopting and following another way that achieves a level of risk that is equal to or better than the acceptable level.

Where a part of a recognised standard or other normative document referred to therein conflicts with the *Coal Mining Safety and Health Act 1999* or the Coal Mining Safety and Health Regulation 2017, the Act or Regulation takes precedence.

Issued under the authority of the Minister for Resources.

Gazetted 22 November 2019

1.0 Foreword

The inquiry into the explosion at the Moura No. 2 Mine in August 1994 recommended - “Emergency procedures should be exercised at each mine on a systematic basis, the minimum requirement being on an annual basis for each mine.”

Five Implementation Task Groups were convened to study the Inquiry recommendations and prepare the mechanisms by which industry could effectively implement them. Task Group 2 was charged with the task of examining and preparing guidelines with respect to emergency procedures exercises. The task was completed in December 1996 with the publication of the “Approved Standard for the Conduct of Emergency Procedures Exercises.”

From 1996 to 1998 significant changes occurred necessitating a review of the Standard, these being:

- Corporatisation of the Queensland Mines Rescue Service (QMRS).
- New legislation with respect to mines rescue requirements in Queensland.
- The acquisition of the GAG Jet Engine Inertisation technology.
- QMRS becoming the custodian and service provider of the GAG Inertisation technology.

The intent of this recognised standard is to provide industry with a set of guidelines that are uniform and consistent. This recognised standard for the conduct of emergency exercises has been developed and reviewed by a cross section of industry stake holders and has taken account of the evolution of opinion currently in the industry in the area of withdrawal of persons and first response in the event of an emergency at a coal mine.

Exercises in withdrawal and first response are a vital part of the safety framework of any coal mine and lead to the opportunity for learning, the sharing of information and opportunity for continuous improvement. Learning from the mistakes of these exercises will encourage all persons employed in the industry to be better prepared should we ever be challenged in real life.

It is only human for us to make mistakes; it is unforgivable not to learn from those mistakes.

Gavin Taylor
Chief Inspector of Coal Mines

2.0 Scope

Moura Recommendations Implementation Task Group 2 was given the following scope regarding the conduct of emergency procedures exercises:

- Guidelines are to be prepared for industry consultation with the objective of conducting well-structured emergency procedures exercises on an annual basis.
- The task group should consider if there is a need for this matter to be covered by legislation, and if so, draft legislation should be prepared for consideration.

3.0 Terms of reference

The guidelines proposed should:

- be systematic;
- be consistent with the concept of mutual assistance from other mines;
- require direct reference to the risks at a mine;
- recognise that exercises should be held not necessarily on day shift;
- be inclusive of external agencies such as QMRS, police, media and senior company officials;
- have an audit and evaluation process; and
- be subject to risk assessment principles to ensure the exercises do not introduce new safety risks to persons at a mine.

As part of the annual exercise and where the scenario requires, consideration should be made as to requiring inertisation equipment to be put in place as well as confirming airlocks and emergency seals on the surface are found to be safely accessible and operative.

4.0 Objectives of emergency exercises

- Safely test the facilities and strategies in place at a mine to manage emergency events in all circumstances.
- Test the competency of mineworkers in using those facilities and implementing the strategies.
- Enhance the confidence and ability of mineworkers to respond in an emergency.
- Identify opportunities for improvement.
- Share the learning outcomes with industry.

4.1 Objectives of this standard

The objectives of this standard are to ensure that Emergency Procedures Exercises are conducted in a manner to:

- ensure there is no personal injury, equipment damage or introduction of additional risks. Note: Design of the emergency exercises must be done using risk assessment methods;
- test the ability of the current Mine Emergency Procedures Plan to meet the desired outcomes of an emergency response;
- relate to the principal hazards identified as being integral to the mine itself and ensure the facilities to control are adequate;
- demonstrate a coordinated response;
- assess all the elements and personnel involved and identify any additional training needs;
- avoid any community alarm / apprehension;
- enhance the confidence and ability to respond to an emergency;
- involve all shifts at some stage through the year. The intent is that the emergency preparedness of the mine is tested for any time of the day or night;
- allow for a performance analysis and debrief to occur with outcomes recorded and relevant information disseminated internally and to the industry. This is to be a formal process in the case of the Level 1 Exercise;

- test the ability of external agencies to respond to an emergency;
- ensure reports and outcomes of all level 2 emergency exercises are sent to the inspectorate in the region for the mine and the Industry Safety and Health Representative (ISHR) by the organising committee within two months of conduct of the exercise;
- ensure the mine forwards any outcomes relevant to emergency response to the inspector for the mine and the ISHR, within six months, an action plan for corrective actions / review of the mines safety and health management system as a result of. all emergency exercises (level 1 - 4); and
- ensure the inspectorate and ISHRs follow up on such plans and ensure that corrective actions that are of benefit to other mines, are circulated in a timely manner.

5.0 Hierarchy of exercise types

A hierarchy of exercise types has been identified comprising four levels and these have been incorporated into a schedule as summarised in Table 1. Implementation of the schedule will:

- assist mines in ensuring that the objectives of emergency procedures exercises are achieved with respect to their operation; and
- provide regular training for QMRS and testing of its ability to provide rescue and inertisation services according to their performance criteria.

The four levels of the exercises are:

- Level 1 – State Level Exercise
- Level 2 – Major Mine Site Exercise
- Level 3 – Minor Mine Site Exercises
- Level 4 – Supporting Exercises

5.1 Level 1 – State level exercise

One mine will be selected each year to respond to an emergency exercise designed and organised by a State Emergency Exercise Executive Management Committee convened under the auspices of the Chief Inspector of Coal Mines. Selection of the mine will be rotated, depending on the life expectance of mines ensuring, where possible, that all mines have at least one level 1 emergency exercise during their life. The mine selected will not be required to undertake the level 2 exercise that year.

5.1.1 Objective and scope

The objectives of this practical exercise are to:

- test the mines emergency response system;
- test the ability of external services to administer assistance; and
- provide a focal point for emergency preparedness in the state.

To this end, the scope of the exercise will include:

- mine response to the scenario presented testing self-escape / aided escape and in-seam response as required; and
- mobilisation of QMRS and other external services, including mines inspectorate, ISHRs, Simtars, police, ambulance to the extent required by the exercise scenario.

QMRS will be expected to:

- provide the rescue team response as defined in the Mines Rescue Agreement with the mine; and
- deploy rescue teams underground including the establishment of a Fresh Air Base (if required) in response to the scenario.

5.1.2 State Emergency Exercise Management Committee

It is the responsibility of the Chief Inspector of Coal Mines to ensure that a State Emergency Exercise Executive Management Committee is convened each year to design, organise, implement and audit the exercise. The Chief Inspector of Coal Mines will determine who is to chair the committee. The Committee is to include:

- one representative of the inspectorate;
- one representative from Simtars;
- three representatives from mines including at least one mine manager;
- one representative from the host mine;
- one representative from QMRS;
- one ISHR; and
- others as may be determined by the above e.g. external experts or service providers.

See Guidelines for the organisation, management and conduct of a Level 1 mine emergency exercise.

5.1.3 Auditing

Performance of the exercise will be audited by a team comprising members of the organising committee and such other personnel as the committee deems necessary. Performance of all key personnel, both internal and external, and the effectiveness of emergency response systems will be assessed. A report detailing the pertinent outcomes of the audit will be disseminated to industry.

5.2 Level 2 – Major mine site exercise

Each year every mine is required to conduct a major practical exercise designed and organised by a committee convened under the auspices of the Site Senior Executive (SSE). It is the principal opportunity for each mine to subject its emergency preparedness to practical scrutiny.

5.2.1 Objective and scope

The objective of the exercise is to test:

- the mines emergency management system;

- communication with external services; and
- periodic mobilisation of the QMRS Inertisation Unit.

To this end the scope of the exercise will include:

- mine response to the scenario presented testing self-escape / aided escape and in-seam response as required;
- communication with QMRS to determine their ability to respond; and
- communication with other external services, as directed by the exercise plan, to determine their ability to respond.

The timing, personnel involved, and scenario of the exercise is to be varied each year to progressively test all aspects of the Emergency Management System.

Notification as part of the exercise shall be made to an Inspector of Mines and ISHRs.

5.2.2 Organising committee

The SSE is responsible for convening this organising committee each year. It is not necessary for the manager to be a member of the committee. Indeed, as the manager normally assumes the predominant role in an emergency, it would often be in the best interests of the exercise that they are not involved.

The committee is to include the most appropriate cross section for the incident and may include a representative from QMRS and a Site Safety and Health Representative.

For example, the SSE may elect to keep the committee “in-house” or to place the task with an external provider or a combination of the two depending on his overall objective.

5.2.3 Auditing

The Organising Committee is to assess the performance of the exercise and of key personnel to identify opportunities for improvement and training needs. The committee may employ additional personnel to achieve this task. A report on the performance and outcomes of the exercise is to be provided to the Regional Inspector of Mines and ISHR, within two months of conduct of the exercise. The Regional Inspector shall ensure that pertinent issues are made available to industry.

5.3 Level 3 – Minor mine site exercise

These practical exercises are to be designed and organised by a committee convened under the auspices of the SSE and conducted during the year. The exercises will progressively test all coal mine workers. Each exercise will test a part of the mine such that during the year all parts of the mine have been progressively tested.

5.3.1 Objective and scope

The objectives of these “incremental” exercises are to:

- to ensure all personnel are familiar with the mine’s Emergency Response Principal Hazard Management Plan (PHMP); and
- provide practical assessment in emergency response including evacuation.

To this end the scope of the exercises will include:

- evacuation of parts of the mine utilising, to the extent practicable, all the facilities and strategies for “self-escape”;
- progressive involvement of all crews during the year i.e. each crew to undertake at least one emergency evacuation each year; and
- involvement of any crew when their work location changes significantly (this can take the form of a familiarisation of the emergency response facilities within the panel, if the donning of the Self-Contained Self Rescuer and an associated changeover is performed this would count as a level 3 exercise). This is to be done as soon as practicable after the change.

NOTE: Those personnel who have participated in an emergency evacuation of the mine as a part of the Level 1 or Level 2 exercises are deemed to have satisfied the requirements of this exercise.

5.3.2 Organising committee

The Underground Mine Manager (UMM) is responsible for organisation, execution and review of the exercises.

5.3.3 Auditing

The UMM shall assess the performance of the exercise and identify opportunities for improvement and training needs. Details and outcomes of the exercise are to be recorded as a mine record entry.

5.4 Level 4 – Supporting exercises

These desktop / semi-practical exercises are to be designed and organised by person(s) under the auspices of the SSE.

5.4.1 Objectives

The objectives of the exercises are to:

- provide theoretical/practical training in emergency response ranging from Incident Management Team (IMT) through to panel scenarios;
- test the ability of personnel to troubleshoot;
- focus on working as a team, test communication and team interaction;
- test knowledge of emergency capability, facilities and systems e.g. medical emergency, ability to perform search and rescue; and
- gain knowledge of potential hazards and causes.

5.4.2 Organising committee

The SSE is responsible for the implementation of the schedule of exercises. The organising person(s) is to ensure the performance of the exercise and of key personnel is assessed as to identify opportunities for improvement and training needs. Details and outcomes of the exercise are to be recorded.

	LEVEL 1 - STATE LEVEL EXERCISE	LEVEL 2 – MAJOR MINE SITE EXERCISE	LEVEL 3 – MINOR MINE SITEVEXERCISE	LEVEL 4 – SUPPORT EXERCISES
TYPE	One mine selected for the sate each year	Whole of mine	Part mine or system	Desktop / semi practical
FREQUENCY	Once per year – at a surface mine every fourth year	Once per year – all mines	Once per year for each crew	Annually / Periodically
SCOPE	Practical exercise to test the emergency response system AND the ability of external services to administer assistance. Involves • Mine response to scenario to self-escape / aided escape and IMT response. • Mobilisation of: ○ QMRS - rescue team response to mines rescue agreement standard – 1 team to be deployed underground. ○ Simtars as required by scenario. ○ External assistance per exercise plan. ****Replaces Level 2 Exercise****	Practical exercise to test the emergency response system including effective communication with external services and periodical mobilisation of the QMRS Inertisation Unit. Involves • Whole mine response. • Mines rescue stations and other external providers to contact stage only – “can you respond?” Each year the day of the week, the time of day and personnel involved is to be verified. The scenario is to be verified each year in order to test all aspects of the mine emergency procedures plan.	Practical test to ensure all personnel are familiar with the Mine Emergency Response or Evacuation Plan. Involves • Part mine evacuation. • All crews. All shifts including weekends. • Whenever a crew’s workplace changes significantly. Participation in Level 1 and 2 exercises qualifies for this exercise.	Desktop/semi practical to test ability to: • respond to a medical emergency. • search and rescue. • provide theoretical / practical training in emergency response including evacuation.
CONTROL	Chief Inspector must ensure the exercise is organised each year.	Chief Inspector must ensure the exercise is organised each year.	SSEs must ensure these exercises are organised.	SSEs must ensure these exercises are organised.

	LEVEL 1 - STATE LEVEL EXERCISE	LEVEL 2 – MAJOR MINE SITE EXERCISE	LEVEL 3 – MINOR MINE SITEVEXERCISE	LEVEL 4 – SUPPORT EXERCISES
	State Emergency Exercise Executive Committee to include one representative from: • Inspectorate • Simtars • Host mine • QMRS • ISHR	Organising Committee to include: • QMRS representative • Site Safety and Health representative	Organising Committee to include: • Site QMRS representative • Site Safety and Health representative	
AUDIT / REPORT	Statewide	Regional Inspector of Mines Industry as required	Regional Inspector of Mines Industry as required	Mine

TABLE 1 – EMERGENCY EXERCISE STANDARD SUMMARY

6.0 Definitions and abbreviations

Trigger Action Response Plans (TARPs) – Elements or sub-elements of an Emergency Response PHMP which are predetermined actions tied to specific trigger levels or events.

Aided Escape – Processes whereby surface personnel are directly involved in assisting underground persons to escape from a mine in an emergency (i.e. information on possible routes to take).

Assisted – An emergency response that requires a mine to access external aid.

Audit – Systematic assessment of performance against written plans, procedures and relevant legislation.

Desktop – An exercise type that is designed to test systems preparedness without involving an attendance or action response i.e. a paper, communication and acknowledgement exercise.

Evacuation – The withdrawal of persons to a place of safety.

Emergency Principal Hazard Management Plan – A management plan developed through the process of risk assessment aimed at ensuring that indicators of principal hazards are identified, monitored and appropriately responded to in a coordinated and orderly manner. Emergency Principal Hazard Management Plans shall:

- be consistent with guidelines established for safety management plans;
- be a single document that systematically defines all actions necessary to ensure that withdrawal of persons to a place of safety are conducted safely;
- include, but not be limited to, organisational structures for emergency response, planning, activities, responsibilities, communications, practices, risks identified, audits and reviews;
- be supplementary to, and include the implementation of appropriate codes, rules, regulations or procedures and is the method by which the withdrawal of persons is to be effectively managed and coordinated; and
- be consistent with the mine emergency procedures and may form part of the mine emergency procedures.

More details on the requirements for a Mines Emergency response plan are included in the Reference material.

Incident Management Team (IMT) – A group of persons with authority defined by a PHMP to initiate actions ensure the safety of mine personnel and to assist with the management of an emergency.

In Seam Response – Response team from an unaffected area of the mine which renders assistance/responds to an underground incident.

Place of Safety – A designated place where persons will assemble without being in any danger from the hazard that triggered the evacuation. The place of safety:

- must reflect the consequence of the hazard that has initiated the evacuation.
- must have an effective means of communication with the surface control.
- may include, but is not limited to, the following locations –
    - Panel crib room.
    - Main headings opposite a district ventilation split.
    - Pit bottom or the base of intake shaft or drift.
    - Refuge chambers / change over stations.
    - Surface location.

Practical Exercise – Realistic simulated scenarios, relevant to principal hazards associated with the mine, and with observations and actions taken to respond to the scenario.

Principal Hazards – Source of potential harm or a situation with a potential to result in multiple fatalities.

Risk Assessment – The process used to determine risk management priorities by evaluating and comparing the level of risk against predetermined standards, target risk levels or other criteria.

Self-Escape – The process of a person(s) escaping from a mine in an emergency without direct assistance from surface personnel.

Self-Reliant – Services, personnel or infrastructure provided and available at the mine site capable of responding to an emergency incident.

Stakeholders – Any party with an interest in the safe operation of the mine.

Surface Control – A competent person on the surface with the authority to initiate and monitor withdrawal of persons to a place of safety.

Trigger Level – A condition / event that is not the normal, is able to be measured or observed, and on being reached requires initiation of predetermined actions.

Withdrawal of Persons – The organised evacuation of persons from the mine or part of the mine to a designated place of safety when the risk to life or health has exceeded predetermined trigger levels.

7.0 Guidelines for the organisation, management and conduct of a Level 1 mine emergency exercise

7.1 Concept

One mine will be selected each year to respond to an emergency exercise designed and organised by a State Emergency Exercise Executive Committee convened under the auspices of the Chief Inspector of Coal Mines.

It was considered that one such exercise per year would achieve the objectives and was sufficient impost on the resources of all those involved.

7.2 Objectives

The objectives of this exercise are to:

- test the mines emergency response system.
- test the mines rescue capability and ability of external services to administer assistance.
- provide a focal point for emergency preparedness in the state.

- form part of the annual audit of performance of accredited corporations for mines rescue (Part 13 of the *Coal Mining Safety and Health Act 1999*)

7.3 Constitution of the State Emergency Exercise Executive Committee

It is the responsibility of the Chief Inspector of Coal Mines to ensure that a State Executive Committee is convened each year to design, organise and implement the exercise. The committee is to include, as a minimum:

- one representative of the inspectorate;
- one representative from Simtars;
- three representatives from mines including at least one mine manager invited by the management committee;
- one representative of QMRS;
- one representative of the ISHRs; and
- others as may be determined by the above e.g. external experts or service providers.

No member of the committee may take part in the mine's or any external agency's response to the emergency procedures exercise.

Committee members will be required to participate in pre-exercise meetings, exercise preparation, and provide assistance in writing relevant sections of the report.

It is anticipated that the delegate from the host mine will play a pivotal role in the planning and conduct of the emergency procedures exercise and, therefore, should be familiar with all operational aspects of the mine including short- and long-term planning.

7.4 Functions of Executive Committee Chairman

The functions of the Executive Committee Chairman would include:

- coordination of the activities of the committee;
- reporting to the Chief Inspector of Coal Mines;
- consultation with senior personnel of any external service providers who are to participate in the exercise. The general nature and objectives of the exercise should be explained and the extent of the commitment to the exercise to be agreed upon;
- consultation with senior company personnel. Agreement should be reached as to the degree of disruption to the operation;
- a two-year plan or schedule should be maintained. The schedule should be made known to industry. This will give ample notice to individual mines as to which year they are scheduled for the Level 1 exercise and to adjust their own schedule of exercises accordingly;
- responsibility for any cancellation or rescheduling of the exercise; and
- production of a report on the exercise and distribution to industry.

7.5 Functions of the Executive Committee

7.5.1 Location selection

Considerations for the selection of the target mine:

- As the level 1 exercise includes the testing of external providers, an analysis of service providers in the different districts should be made.
- Excessive impact on individual external agencies should be avoided. This may be achieved by varying the areas and / or the nature of the exercise on a yearly basis.
- The mines and operating companies involved should be appropriately varied each year where possible.
- Where emerging issues are noted, the mine selection may need to change to test new ideology / technology.

7.5.2 Planning

- All aspects of the plan are to be based on the risk management process.
- The Committee must develop the exercise in accordance with this standard.
- The exercise should simulate as far as practicable actual emergency conditions e.g. use of smoke glasses to simulate minimum visibility; use of training or real self-rescuers where possible and available.
- A thorough review of the host mine should be undertaken prior to the development of the specific exercise. The review should include:
    - an overview of operation and underground inspection where appropriate.
    - determination of principal hazards identified by the safety management plans.
    - emergency response plans.
    - a history of emergency exercises previously conducted.
- The “exercise only” status must be effectively communicated to the families of mine personnel and the community at large to ensure anxieties are not created while also maintaining an element of surprise. This can be achieved by revealing to the mine and the community, at an appropriate time prior to the exercise, a time period in which the exercise will be initiated e.g. a “window” of one or two weeks.
- Consider the involvement of the media.
- Consideration must be given to the operational and safety status of the mine prior to the initiation of the emergency procedures exercise. It is a function of the host mine delegate to provide timely information to the committee chairman, which would cause the committee to alter or re-schedule the exercise.

7.5.3 Auditing

- An audit plan will need to be developed for the exercise which will require:
    - a detailed review of the mines overall “emergency procedures plan” or “emergency response plan” and the “mine evacuation plan” or “emergency evacuation plan” which sits beneath it; and
    - a detailed review of the QMRS procedures and performance.
- Identify the number of assessors required.
- Recruit additional assessors as required.
- Allocate assessment functions to members of the committee and others.
- Determination of the effectiveness of the plan to facilitate continuous improvement.

7.5.4 Key issues for auditing

Note: Some of the identified issues may not be applicable for Level 1 exercises at surface mines

- Immediate Response
    - Preservation of life.
    - Minimise potential harm to persons.
    - Prevent deterioration of conditions.
    - Promote recovery.
- Effectiveness of the Incident Management Team
    - Incident initiation.
    - Establishing the team.
    - Accommodation and facilities.
    - Information transfer.
    - Information processing – including debriefs and flow of information in a timely manner.
    - Ventilation evaluation and gas analysis.
    - Decision making process.
    - Risk assessment process.
    - Availability of plans.
    - Use of external experts.
    - Communications to mine recovery team(s) / QMRS.
    - Social/HR information and issues.
    - Briefing of corporate incident team.
    - IMT develop the desired outcomes of the Incident.
    - Span of control.
    - Safety, health and wellbeing.
    - Delegate responsibility.
    - Effect changeovers.
    - Information recording.
- Effectiveness of the Mine Control Room
    - Information content and flow – in and out of control room.
    - Event log keeping.
    - Availability of competent monitoring / interpreting personnel.
    - Verification of missing persons.
    - Location of persons.
    - Monitoring of ventilation.
    - Emergency communications.
    - Availability of plans.
    - Documented action plan.
    - Use of duty cards.
    - Hazards identification and risk assessment.
    - Gas monitoring.
- With Respect to “SELF ESCAPE” Facilities and Strategies
    - Ability of personnel to don SCSRs or other apparatus.
    - Performance of SCSRs – a sample usage.
    - Location and use of caches.

  - Change over stations.
  - Alternate escape routes.
  - Knowledge of:
    - effectiveness of signage and other aids.
    - contact time.
    - control of personnel location.
    - availability of plans.
    - machinery (transport).
    - Communications.
- With Respect to “AIDED ESCAPE” Facilities and Strategies
  - Location of persons determined.
  - Contract made with persons.
  - Arrival of external assistance (QMRS, Simtars, etc).
  - Arrival time.
  - Organisation of QMRS sub station.
  - Availability of plans etc.
  - Effectiveness of briefing.
  - Efficiency of preparation and time to go underground.
  - Compliance with designated underground task.
  - Compliance with SOP and Mines Rescue Guidelines.
  - Team reporting and debriefing.
  - Deployment time of teams.

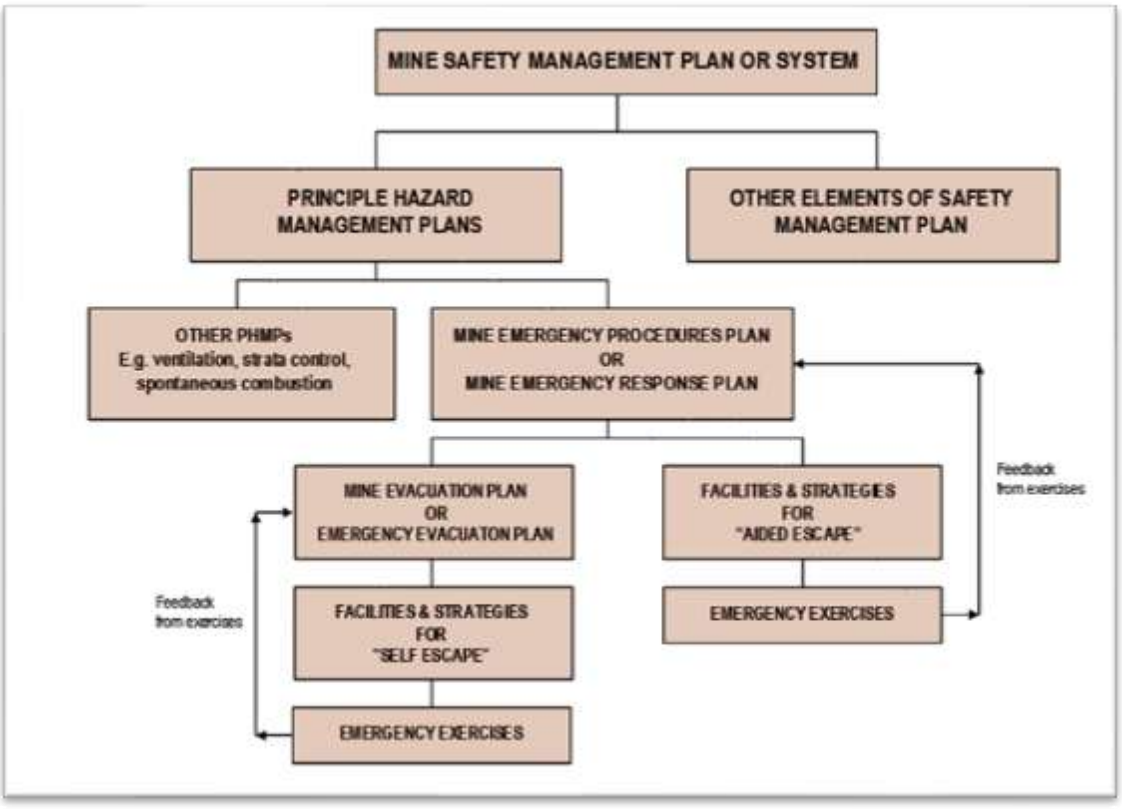
8.0 Mine emergency response plan

The Mine Emergency Response Plan is to be audited / reviewed both internally and externally. The following basic audit criteria will assist in that process (may not be applicable to all types of mines).

PLAN ELEMENT	AUDIT / REVIEW CRITERIA	FREQUENCY
1. Identify principal hazards	System to identify key hazards.	Annual or changed circumstances.
2. Define roles and responsibilities	Compliance with Mine Emergency Response Plan.	As above.
3. Competency in roles and responsibilities	Competency based assessment / challenge test.	On appointment and to be defined by Mine Emergency Response Plan or other safety management plan.
4. Familiarity with mine egress routes, ventilation systems, communication systems, emergency equipment	As above.	As above.
5. Schedule of audits	Audit review process.	As defined for each element of Mine Emergency Response Plan or other safety management plan.
6. On-site facilities	Mine plans and documented checklists.	As above.

PLAN ELEMENT	AUDIT / REVIEW CRITERIA	FREQUENCY
7. Defined command structure	Evidence of challenge test undertaken.	Annual minimum or as determined by Mine Emergency Response Plan or other safety management plan.
8. Communication procedures • Hardware • Information flow • Information recording	• Documented inspection and testing. • Documented evidence of challenge test.	As above.
9. Specific event response relating to key hazards	As above.	As above.
10. Documented control	Evidence of document control.	As determined by Mine Emergency Response Plan or other safety management plan.
11. Identify resourcing and integrating of external services	Documented evidence of challenge test.	As above.
12. Emergency response equipment and installations e.g surface airlocks / emergency seals	Operable and accessible.	As determined by Mine Emergency Response Plan or other safety management plan.

9.0 Safety management plans hierarchy and link to emergency exercises



10.0 Reference material

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